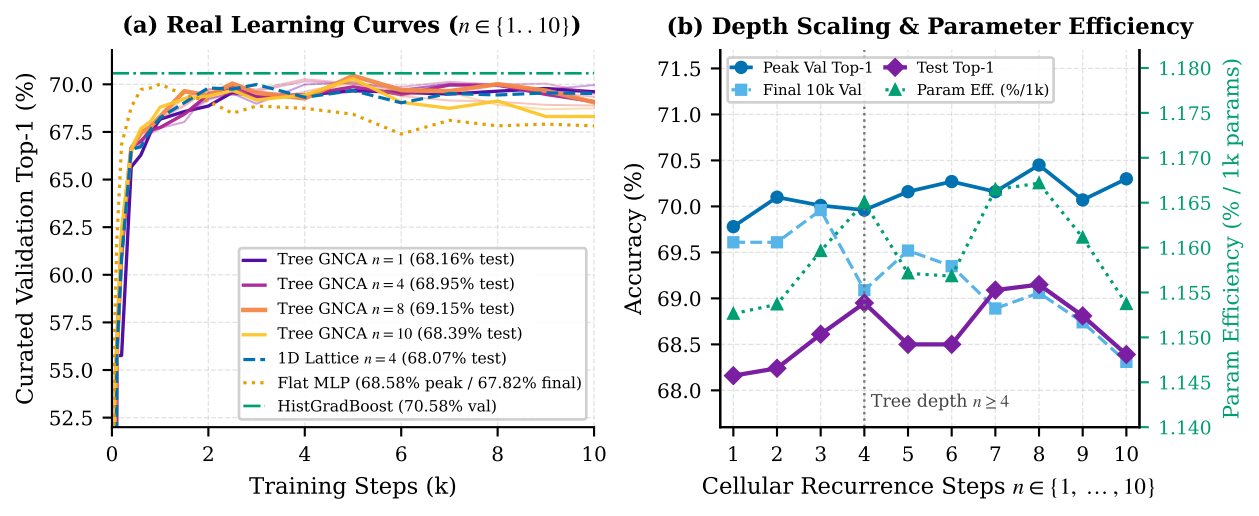
(a) Real Learning Curves ($n \in \{1..10\}$)
Curated Validation Top-1 (%)
Training Steps (k)
Tree GNCA $n=1$ (68.16% test)
Tree GNCA $n=4$ (68.95% test)
Tree GNCA $n=8$ (69.15% test)
Tree GNCA $n=10$ (68.39% test)
1D Lattice $n=4$ (68.07% test)
Flat MLP (68.58% peak / 67.82% final)
HistGradBoost (70.58% val)
(b) Depth Scaling & Parameter Efficiency
Peak Val Top-1
Test Top-1
Final 10k Val
Param Eff. (%/1k)
Accuracy (%)
Param Efficiency (% / 1k params)
Tree depth $n \geq 4$
Cellular Recurrence Steps $n \in \{1, \ldots, 10\}$